

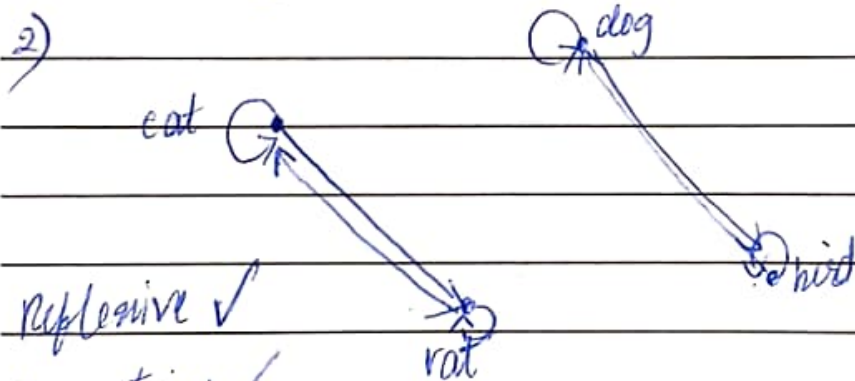
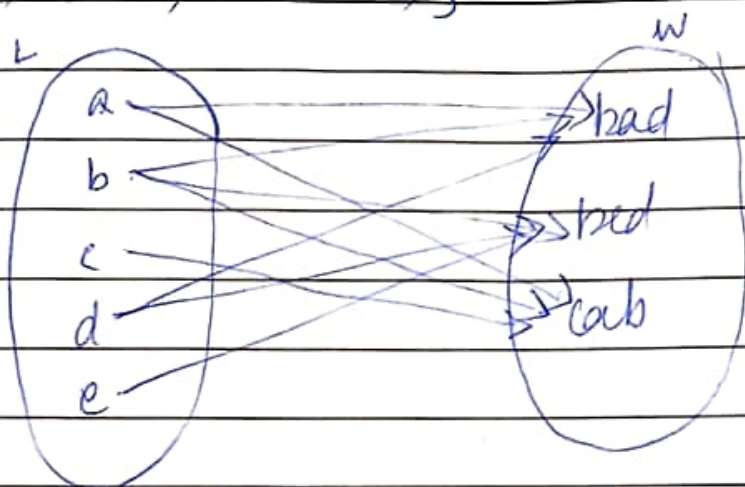
# Mergestack

4.03  
pg 196

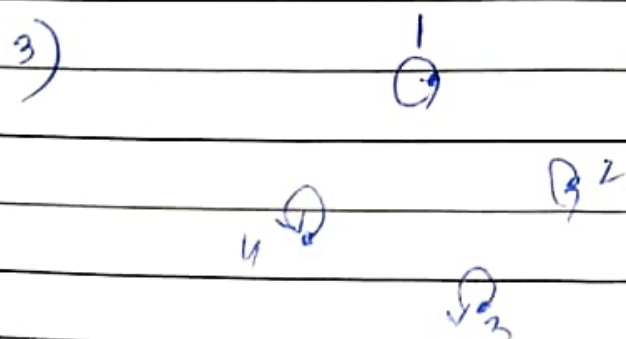
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1)  $R: \{ (a, bad), (a, cab), (b, bad), (b, hed), (b, cab), (c, cab), (d, bad), (d, hed), (e, bed) \}$



Reflexive ✓  
Symmetric ✓  
Transitive X



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- 4) (a)  $\{(a, c), (b, d), (c, c), (d, a), (d, b)\}$   
 (b)  $\{(a, b), (a, d), (b, a), (b, d)\}$   
 (c)  $\{(a, a), (b, b), (b, d), (c, c), (d, d), (d, d)\}$   
 (d)  $\{(a, b), (a, c), (a, d), (b, d), (c, d), \}$
- 5)  $S \circ R = \{(a, y), (a, x), (b, x), (c, y), (c, x)\}$

6)  $D_R \circ D_S = D_{R+S}$

$\rightarrow \text{Supp}(a, c) \in D_{R+S}$

$(a, c) \in D_{R+S}$

$\exists (a, b) \in D_S \ \& \ (b, c) \in D_R$

$|a - c| < r + s$

all 3 of these are

$|a - c| < r + s$   
 $|a - b| < s$   
 $|b - c| < r$

We know that  $(a - c) = (a - b) + (b - c)$

so  $|a - c| = |(a - b) + (b - c)|$

Using Triangle inequality  $|x + y| \leq |x| + |y|$

so  $|a - c| = |(a - b) + (b - c)| \leq |a - b| + |b - c| < s + r$   
 as  $|a - b| < s$   
 &  $|b - c| < r$   
 so  $|a - c| < s + r$

$\leftarrow \text{supp } |a - c| < r + s$

$(a, c) \in D_{R+S}$

$\exists b [ |a - b| < s \ \& \ (b - c) < r ]$

Let  $b = \frac{ra - sc}{r + s}$

This is actually found by weighted mean geometry

so  $|a - b| = \left| a - \frac{ra - sc}{r + s} \right| = \left| \frac{ar + as - ar - sc}{r + s} \right| = \left| \frac{as - sc}{r + s} \right|$

$= \frac{s}{r + s} |a - c| < \frac{s}{r + s} (r + s) = s$  so  $|a - b| < s$

$\therefore (as |a - c| < r + s)$

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$$\begin{aligned} 8 \mid b-c &= \left| \frac{na-sc}{n+s} - c \right| = \left| \frac{na-sc-cr-cs}{n+s} \right| = \left| \frac{na-nc}{n+s} \right| \\ &= \frac{n|a-c|}{n+s} < \frac{n}{n+s} (n+s) = n \text{ so } (b-c) < n. \end{aligned}$$

7)  $R$  is reflexive.  $\iff \forall x \in A (x, x) \in R$   $\iff \text{if } x \in A \text{ then } (x, x) \in R$  (see pg 33)

$R \subseteq A \times A$

$\forall x \in A \{ (x, x) \in R \} \subseteq R$

$\forall x \in A \{ (x, x) \in R \} \rightarrow (x, x) \in R$

$\forall x \in A \{ (x, x) \in R \} \rightarrow (x, x) \in R$

$\rightarrow$  Suppose  $x \in A$  &  $(x, x) \in A$ . Since  $R$  is reflexive so  $(x, x) \in R$ . so  $ia \subseteq R$ .

If we take  $(x, y) \in A$  &  $x=y$  then since  $R$  is reflexive  $(x, x) \in R$  as  $x=y \in A$ . putting  $y=x$   $(x, x) \in R$ .

$\leftarrow$

Supp  $\forall (x, y) \in A$

$$\forall x \in A ((x, x) \in A) \rightarrow (x, x) \in R \quad \forall x \in A ((x, x) \in R)$$

Supp  $x \in A$

so  $(x, x) \in A$

$(x, x) \in R$

exists  $ia \subseteq R$  so  $(x, x) \in R$  so  $R$  is reflexive

$$\forall (x, y) \in A \times A ((x, y) \in A \rightarrow (x, y) \in R)$$

8)

$R$  is transitive

$\iff$

$R \circ R \subseteq R$

$$\forall x \in R \forall y \in R \forall z \in R ((x, y) \in R \wedge (y, z) \in R) \rightarrow (x, z) \in R$$

Supp  $x, y, z \in R$  then  $\forall x, z \exists y ((x, y) \in R \wedge (y, z) \in R) \rightarrow (x, z) \in R$   
 goes easily follows



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$$\leftarrow \text{supp } xRy \wedge yRz \quad xRz$$

Then since we have  $y$  such that

$$xRy \text{ \& } yRz \text{ so } xRz \text{ as } (x,z) \in R \circ R$$

$$\text{ \& } R \circ R \subseteq R \text{ so } (x,z) \in R \text{ for arbitrary } x \text{ \& } z$$

$$9) R \subseteq A \times B$$

$$\rightarrow R \circ i_A = R$$

$$\text{since } i_A \subseteq A \times A \quad (a,b) \in R$$

$$\text{so } R \circ i_A \subseteq A \times B$$

$$\text{supp } (a,b) \in R \circ i_A$$

$$\text{iff } \exists a' \in A [(a,a') \in i_A \wedge (a',b) \in R]$$

$$\text{since } (a,a') \in R \circ i_A \text{ \& } (a',b) \in R \text{ so } R \circ i_A \subseteq R.$$

$$\leftarrow \text{since } (a,a') \in i_A \text{ so } a=a' \text{ so } (a',b) = (a,b) \in R.$$

$$\text{supp } (a,b) \in R \quad (a,b) \in R \circ i_A$$

$$\exists a' \in A [(a,a') \in i_A \wedge (a',b) \in R]$$

Let  $a = a' \in A$ . By definition of  $i_A$

$$(a,a') \text{ must be in } i_A \text{ \& since } (a',b) \in R$$

$$\text{so } (a,b) \in R \circ i_A \text{ so } R \subseteq R \circ i_A$$

$$(b) i_B \circ R = R$$

$$\text{since } R \subseteq A \times B$$

$$i_B \subseteq B \times B$$

$$\text{so } i_B \circ R \subseteq A \times B$$

$$\text{supp } (a,b) \in i_B \circ R \quad (a,b) \in R$$

$$\exists b' \in B [(a,b') \in R \wedge (b',b) \in i_B]$$

$$\text{since } (b',b) \in i_B \text{ so } b'=b \text{ so } (a,b') = (a,b) \in R.$$

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$$\leftarrow \text{supp } (a,b) \in R \quad \exists b \in B [(a,b) \in R \wedge (b,b) \in R]$$

we already know that  $(b,b) \in R$   
for every  $b \in B$ . so goal follows.

10)  $S \subseteq A \times A$

$$D = \text{Dom}(S) = \{a \mid \exists b \in B (a,b) \in S\}$$

$$R = \text{Ran}(S) = \{b \mid \exists a \in A (a,b) \in S\}$$

$$\text{id} \subseteq S^{-1} \circ S \Rightarrow \forall (a,b) \in A \times A [(a,b) \in \text{id} \rightarrow (a,b) \in S^{-1} \circ S]$$

• Note that  $\text{id} \subseteq A \times A$

$$\text{supp } (a,b) \in \text{id}$$

$$(a,b) \in S^{-1} \circ S$$

$$\text{iff } a \in D \wedge b \in D \wedge a = b$$

$$\exists b \in B [(a,b) \in S^{-1} \wedge (b,a) \in S]$$

$$\text{since } a \in D \text{ so } \exists b \in A (a,b) \in S$$

$$'' \quad '' \quad \wedge (a,b) \in S$$

• since  $(a,b) \in S$  so goal follows

(b)  $\text{id} \subseteq S \circ S^{-1}$

$$\text{supp } (a,b) \in \text{id}$$

$$(a,b) \in S \circ S^{-1}$$

$$\text{so } a = b \wedge b \in R$$

$$\exists a \in A [(b,a) \in S^{-1} \wedge (a,b) \in S]$$

$$\text{since } b \in R \text{ so } \exists a \in A (a,b) \in S$$

$$\exists a \in A [(a,b) \in S \wedge (a,b) \in S]$$

$$\text{so } (a,b) \in S \wedge (b,a) \in S^{-1}$$

$$\text{so we found } a = a$$

↓  
in both of these, note  $a=b$ , so  $(b,b) = (a,a) \in S \circ S^{-1}$  so goal is proved  
just by plugging in values



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1)  $R \subseteq A \times A$

$R$  is reflexive

$$\forall x \in A (x, x) \in R$$

$$\text{Supp } (x, y) \in R$$

$$R \subseteq R \circ R$$

$$\forall (x, y) \in A \times A \mid (x, y) \in R \rightarrow (x, y) \in R \circ R$$

$$(x, y) \in R \circ R$$

$$\exists z \mid (x, z) \in R \wedge (z, y) \in R$$

Let  $z = x$ , we know  $(x, x) \in R$  as  $R$  is reflexive

& also by assumption  $(x, y) \in R$  so goal follows

12)

$$R \subseteq A \times A$$

(a)  $R$  is reflexive

$$\text{supp } (x, x) \in R$$

$$\forall x \in A (x, x) \in R$$

$R^{-1}$  is reflexive

$$\forall x \in A (x, x) \in R^{-1}$$

$$\forall x \in A (x, x) \in R^{-1}$$

$$(x, x) \in R^{-1} \text{ supp } x \in A. \text{ since } (x, x) \in R \text{ so } (x, x) \in R^{-1}$$

(b)  $R$  is symmetric

$$\forall x \in A \forall y \in A (x R y \rightarrow y R x)$$

$$\text{Supp } y R x \text{ so } (y, x) \in R^{-1}$$

$$\text{so } (x, y) \in R \text{ & since } R \text{ is symmetric}$$

$$(y, x) \in R \text{ so } (x, y) \in R^{-1}. \text{ if } x R^{-1} y \text{ so } R^{-1} \text{ is also symmetric}$$

(c)  $R$  is transitive

$$\forall x \forall y \forall z [(x R y \wedge y R z) \rightarrow x R z] \text{ ①}$$

$$\text{Supp } x R^{-1} y \wedge y R^{-1} z$$

$$\text{so } y R x \wedge z R y$$

$$\text{By ① we get } z R x \text{ so } x R^{-1} z$$

$R^{-1}$  is transitive

$$\forall x \forall y \forall z [(x R^{-1} y \wedge y R^{-1} z) \rightarrow x R^{-1} z]$$

$$\rightarrow x R^{-1} z$$

proven

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$$(a) R_1 \subseteq A \times A \quad R_2 \subseteq A \times A$$

$R_1$  &  $R_2$  are reflexive

$R_1 \cup R_2$  is reflexive?

$$\forall x \in A (x, x) \in R_1$$

$$\forall x \in A (x, x) \in R_1 \cup R_2$$

$$'' '' \in R_2$$

Supp  $x \in A$  so  $(x, x) \in R_1$  and  $(x, x) \in R_2$  since both are reflexive  
so  $(x, x) \in R_1 \cup R_2$

(b)  $R_1$  &  $R_2$  are symmetric

$R_1 \cup R_2$  ?

$$\forall x \in A \forall y \in A [x R_1 y \rightarrow y R_1 x]$$

$$\forall x \in A \forall y \in A [x R_1 \cup R_2 y \rightarrow y R_1 \cup R_2 x]$$

$$'' '' [x R_2 y \rightarrow y R_2 x]$$

$$(x, y) \in R_1 \cup R_2 \rightarrow$$

$$(y, x) \in R_1 \cup R_2$$

supp  $x \in A, y \in A$  &  $(x, y) \in R_1 \cup R_2$

so  $(x, y) \in R_1 \vee (x, y) \in R_2$

Case 1)  $(x, y) \in R_1$ . since  $R_1$  is symmetric so  $(y, x) \in R_1$ . so  $(y, x) \in R_1 \cup R_2$ .

Case 2) similar

(c)  $R_1$  &  $R_2$  are transitive

$R_1 \cup R_2$  ?

No, it's not.

$$\text{supp } \forall x \in A \forall y \in A \forall z \in A [(x R_1 y \wedge y R_1 z) \rightarrow x R_1 z]$$

& similar for  $R_2$

$$\forall x \in A \forall y \in A \forall z \in A$$

$$[(x, y) \in R_1 \cup R_2 \wedge (y, z) \in R_1 \cup R_2] \rightarrow (x, z) \in R_1 \cup R_2$$



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let  $x, y, z$  be arbitrary & suppose

$$(x, y) \in R_1 \cup R_2$$

$$(x, z) \in R_1 \cup R_2$$

$$(y, z) \in R_1 \cup R_2$$

All cases don't follow. e.g.

Case 1) Follows  $(x, y) \in R_1, (y, z) \in R_1$  then since  $R_1$  is transitive so  $(x, z) \in R_1$  & so  $(x, z) \in R_1 \cup R_2$

But case 2 doesn't follow:  $(x, y) \in R_1, (y, z) \in R_2$

Counterexample:  $A = \{1, 2, 3\}$   
 $R_1 = \{(1, 2), (2, 3), (1, 3)\}$   
 $R_2 = \{(2, 2), (2, 3)\}$

Both are transitive

But  $R_1 \cup R_2$  has  $(1, 2)$  &  $(2, 2)$  but not  $(1, 2)$

14)  $R_1$  &  $R_2$  are reflexive

$R_1 \cap R_2$ ?

$$\forall x \in A [(x, x) \in R_1 \cap R_2]$$

Supp  $x \in A$ . since  $R_1$  &  $R_2$  are reflexive

$$\text{so } (x, x) \in R_1 \text{ & } R_2 \text{ so } (x, x) \in R_1 \cap R_2$$

(b)  $R_1$  &  $R_2$  are symmetric

$R_1 \cap R_2$

$$\forall x \in A \forall y \in A [(x, y) \in R_1 \cap R_2 \rightarrow (y, x) \in R_1 \cap R_2]$$

$$\text{supp } (x, y) \in R_1 \cap R_2$$

so  $(x, y) \in R_1$  &  $(x, y) \in R_2$ . Since both are symmetric

$$\text{so } (y, x) \in R_1 \text{ & } (y, x) \in R_2 \text{ so } (y, x) \in R_1 \cap R_2$$



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(c)  $R_1$  &  $R_2$  are transitive

$R_1 \cap R_2$  ?

We have to prove that  $\forall x, y, z \in A$   $\left[ (x, y) \in R_1 \cap R_2 \wedge (y, z) \in R_1 \cap R_2 \right] \rightarrow (x, z) \in R_1 \cap R_2$

supp  $(x, y) \in R_1 \cap R_2$   $(y, z) \in R_1 \cap R_2$   $(x, z) \in R_1 \cap R_2$

since  $(x, y) \in R_1$   $\wedge$   $(y, z) \in R_2$  so  $(x, z) \in R_1 \cap R_2$  so goal follows

(5)  $R_1$  &  $R_2$  are reflexive

$R_1 \setminus R_2$  ?

$\forall x \in A \left[ (x, x) \in R_1 \wedge (x, x) \notin R_2 \right]$

supp  $x \in A$ .  $R_1$  &  $R_2$  are reflexive so  $(x, x) \in R_1 \cap R_2$ .

Consider example

supp  $A = \{1, 2\}$   $R_1 = \{(1, 1), (2, 2)\}$  but  $R_1 \setminus R_2 = \{ \}$ .

(b)  $R_1$  &  $R_2$  are symmetric

$R_1 \setminus R_2$  ?

$\forall x, y \in A \left[ (x, y) \in R_1 \setminus R_2 \rightarrow (y, x) \in R_1 \setminus R_2 \right]$

supp  $(x, y) \in R_1 \setminus R_2$  then

$(x, y) \in R_1 \wedge (x, y) \notin R_2$

• since  $(x, y) \in R_1$  so  $(y, x) \in R_1$ . And since  $(x, y) \notin R_2$  so

$(y, x) \notin R_2$ . Since if  $(y, x) \in R_2$  then  $(x, y)$  must belong

to  $R_2$  since  $R_2$  is symmetric.

so  $(y, x) \in R_1 \setminus R_2$

In  $\forall x, y \left[ (x, y) \in R_2 \rightarrow (y, x) \in R_2 \right]$

We can take  $(y, x)$  for some known  $x, y$  so we can conclude

(c)  $R_1$  &  $R_2$  are transitive

$R_1 \setminus R_2$  ?

$\forall x, y, z \left[ (x, y) \in R_1 \setminus R_2 \wedge (y, z) \in R_1 \setminus R_2 \rightarrow (x, z) \in R_1 \setminus R_2 \right]$

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supp  $(x, y) \in R_1 \setminus R_2$  then  $x$  may  $\notin R_2$   $(x, z) \in R_1 \setminus R_2$   
 $\& (y, z) \in R_1 \setminus R_2$  then  $(y, z) \in R_1 \cap (y, z) \notin R_2$   
 since  $(x, y) \in R_1$   $\& (y, z) \in R_1$  so  $(x, z) \in R_1$   
 but we cannot conclude if  $(x, z) \notin R_2$ . It may or may not.  
 So  $R_1, R_2$  is not always transitive

Counterexample:

$$R_1 = \{(1, 2), (2, 3), (1, 3)\} \quad \text{--- non transitive but } R_1 \setminus R_2 \text{ has } (1, 2) \& (2, 3) \text{ but not } (1, 3)$$

$$R_2 = \{(1, 3), (3, 3)\}$$

16)  $R \subseteq A \times A$   $S \subseteq A \times A$

$R, S$  are reflexive

$$\forall x \in A ((x, x) \in R \wedge (x, x) \in S)$$

supp  $x \in A$

let  $y = x$ . We have  $(x, y) = (x, x) \in R \& S$ . so  $R \cup S$  is reflexive

$R \cup S$  is reflexive

$$\forall x \in A ((x, x) \in R \cup S)$$

$$(x, x) \in R \cup S$$

$$\exists y [(x, y) \in S \wedge (y, x) \in R]$$

17)  $R \& S$  are symmetric relations on  $A$

$R \cup S$  is symmetric iff  $R \cup S = S \cup R$

→ supp  $R \cup S$  is symmetric

$$R \cup S = S \cup R$$

$$\forall (x, y) \in R \cup S \rightarrow (y, x) \in R \cup S$$

→ supp  $(x, y) \in R \cup S$ . Since  $R \cup S$  is symmetric

$$\& (y, x) \in R \cup S$$

$$\text{iff } \exists c [(y, c) \in S \wedge (c, x) \in R]$$

$$(y, y) \in S \cup R$$

$$\exists z [(x, z) \in R \wedge (z, y) \in S]$$

so  $(y, c) \in S \wedge (c, x) \in R$ . Since  $R \& S$  are symmetric

so  $(c, y) \in S \wedge (x, c) \in R$ . so let  $z = c$ . goal follows



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$\leftarrow$  supp  $(x, y) \in SOR$  give  $SOR \cup S$   $(m, y) \in ROS$

symmetric so  $(y, x) \in SOR$ .

$\exists z [(x, z) \in S \wedge (z, y) \in R]$

so  $\exists c [(y, c) \in R \wedge (c, x) \in S]$

since  $R \cup S$  is symmetric.  $(c, y) \in R \wedge (x, c) \in S$ . so taking  $z = c$ , goal follows

$\leftarrow$  supp  $ROS = SOR$

$ROS$  is symmetric

$\forall (x, y) \in A \times A [(x, y) \in ROS \iff (x, y) \in SOR]$  ①  $\forall (x, y) \in A \times A$

supp  $(x, y) \in ROS$

$\{(x, y) \in ROS \rightarrow (y, x) \in ROS\}$

we have ①  $(x, y) \in SOR$

iff  $\exists c [(x, c) \in R \wedge (c, y) \in S]$

$\exists z [(y, z) \in S \wedge (z, x) \in R]$

But since  $R \cup S$  are symmetric

so  $(c, x) \in R \wedge (y, c) \in S$  so let  $z = c$ . Our goal follows

1a)  $R, S \subseteq A \times A$

$\stackrel{so}{(R \subseteq ROS)} \rightarrow ROS$  is transitive

Both are transitive

supp  $R \subseteq ROS$

$ROS$  is transitive

$\forall x \forall y \forall z [(x, y) \in ROS \wedge (y, z) \in ROS \rightarrow (x, z) \in ROS]$

supp  $(x, y) \in ROS \& (y, z) \in ROS$

$\exists a [(x, a) \in R \wedge (a, y) \in R]$

$(x, z) \in ROS$

$\exists b [(y, b) \in S \wedge (b, z) \in R]$

since  $(a, y) \in R \& (y, b) \in S$  so  $(a, b) \in SOR$

since  $R \subseteq ROS$  so  $(a, b) \in ROS$

iff  $\exists c [(a, c) \in S \wedge (c, b) \in R]$

now since  $(a, b) \in S \wedge (a, c) \in S$  so  $(x, c) \in S$  as  $S$  is transitive



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from  $(c,b) \in R \wedge (b,z) \in R$  and  $R$  is transitive, so  $(c,z) \in R$ .

Now finally as  $(x,c) \in S \wedge (c,z) \in R$  so  $(x,z) \in R \circ S$   
so  $R \circ S$  is transitive

19) There is incorrect. The mistake in proof is that in existential instantiation on  $(x,y) \in S \wedge (y,z) \in S$  means both  $x$  and  $y$  can be different. They need not be same.

Counterexample: let  $A = \{1,2\}$   $R = \{(1,2), (2,1), (1,1)\}$   
so  $R$  is transitive. Note that  $(\{2\}, \{1,2\}) \in R$  &  $(\{1,2\}, \{2\}) \in R$   
both belong to  $S$  but  $(\{2\}, \{2\}) \notin S$ .

20)  $R \subseteq A$

$$B = \{x \in P(A) \mid x \neq \emptyset\}$$

$$S = \{(x,y) \in B \times B \mid \forall x \in x \forall y \in y (x R y)\}$$

$R$  is transitive

$S$  is transitive

$$\forall x \in B \forall y \in B \forall z \in B$$

$$((x,y) \in S \wedge (y,z) \in S) \rightarrow (x,z) \in S$$

let  $x, y, z$  be arbitrary

$$\text{supp } (x,y) \in S \wedge (y,z) \in S$$

$$(x,z) \in S$$

$$\forall x \in x \forall y \in y (x R y) \quad \text{--- (1)}$$

$$\forall x \in x \forall z \in z (x R z)$$

$$\forall y \in y \forall z \in z (y R z) \quad \text{--- (2)}$$

$$\text{let } x \in x \text{ \& } z \in z \text{ be arbitrary}$$

$$\text{from } x R y \text{ \& } y R z \text{ for arbitrary } x, y, z \text{ so}$$

$$x R z$$

$$x R z \text{ \& therefore } (x,z) \in S \text{ so } S \text{ is transitive}$$

Since  $y \in B$  so  $y \neq \emptyset$  so we can choose some  $y \in y$ , let  $y \in y$

Now using (1) & (2), we can conclude  $x R y$  &  $y R z$  so

$x R z$  as required

2)  $R \subseteq A \times A$

$$S = \{ (X, Y) \in P(A) \times P(A) \mid \forall x \in X \exists y \in Y (x R y) \}$$

(a)  $R$  is reflexive

$S?$

$$\forall x \in A (x, x) \in R \quad \text{--- ①}$$

$$\forall X \in P(A) [(X, X) \in S]$$

$$\text{Supp } X \in P(A) \Rightarrow X \subseteq A$$

$$(X, X) \in S$$

$$\text{Supp } x \in X$$

$$\forall x \in X \exists y \in X (x R y)$$

$$\text{Since } X \subseteq A \text{ so } x \in A \text{ so by ①}$$

$$\exists y \in X (x R y)$$

$$(x, x) \in R \text{ or } x R x$$

<sup>1</sup> This can be named  $y$  also

So let  $x$  be arbitrary follows

(b)  $R$  is symmetric

$S?$

$$\forall x \in A \forall y \in A [x R y \rightarrow y R x]$$

$$\forall X \in P(A) \forall Y \in P(A)$$

$$\text{Supp } X \in P(A) \text{ so } X \subseteq A$$

$$[(X, X) \in S \rightarrow (Y, X) \in S]$$

$$\& Y \in P(A) \text{ so } Y \subseteq A$$

$$(Y, X) \in S$$

$$\text{Supp } (X, Y) \in S$$

$$\forall y \in Y \exists x \in X (y R x)$$

$$\forall x \in X \exists y \in Y (x R y)$$

$$\text{Supp } y \in Y \text{ for arbitrary } y$$

We cannot conclude here anything for arbitrary  $y$  because we don't know whether  $\forall x (x \in X) \text{ or not}$

Counterexample: Let  $A = \{1, 2\}$   $P = \{\{1, 2\}, \{2, 1\}\}$  so  $R$  is symmetric

Note that  $(\{1\}, \{1, 2\}) \in S$  but  $(\{1, 2\}, \{1\}) \notin S$



# Mergestack

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1)  $R$  is transitive

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$$\forall x \in P(A) \forall y \in P(A) \forall z \in P(A)$$

Let  $x, y, z \in P(A)$  be arbitrary

Supp so  $x \subseteq A, y \subseteq A, z \subseteq A$

$$\text{Supp } (x, y) \in S \wedge (y, z) \in S$$

$$\forall x \in x \exists y \in y (x R y) \quad \text{--- ①}$$

$$\forall y \in y \exists z \in z (y R z) \quad \text{--- ②}$$

Supp  $x \in x$  be arbitrary. As  $x \subseteq A$

so  $x \in A$  also.

by ① we can choose some  $y_1 \in y$  such that  $x R y_1$ .

Also  $y_1 \in A$  as  $y \subseteq A$ .

by ② we can conclude  $y_1 R z_1$  for some  $z_1 \in z$ . Also  $z_1 \in A$  as  $z \subseteq A$ .

Now as  $y_1 R z_1$  &  $x R y_1$ , and all  $x, y, z \in A$  &  $R$  is transitive, we can conclude  $x R z_1$ .

22) Incorrect proof as we don't know any  $y \in A$  exist or not.

Thm is incorrect - Counterexample:  $A = \{1\}, R = \emptyset$ .

$\emptyset$  is indeed  $\subseteq A \times A$  as  $\emptyset \subseteq$  any set.

23)  $F \subseteq P(A)$

$$R = \{(a, b) \in A \times A \mid \forall x \subseteq (A \setminus \{a, b\}) [x \cup \{a\} \in F \rightarrow x \cup \{b\} \in F]\}$$

Prove that  $R$  is transitive

# Mergestack

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$$\forall a, b, c \in A [(aRb \wedge bRc) \rightarrow aRc]$$

Supp  $a, b, c \in A$  be arbitrary

supp  $aRb$  &  $bRc$

$aRc$

$(a, b) \in R, (b, c) \in R$

$$24) R = \{ (m, n) \in \mathbb{N} \times \mathbb{N} \mid |m - n| \leq 1 \} \quad \begin{matrix} R \subseteq \mathbb{N} \times \mathbb{N} \\ R \subseteq \mathbb{Z} \times \mathbb{Z} \end{matrix}$$

(a)  $R$  is reflexive on  $\mathbb{N}$ ?

$$\forall m, n \in \mathbb{N} [(m, n) \in R \wedge (n, m) \in R]$$

$$\forall m \in \mathbb{N} [(m, m) \in R]$$

Supp  $m \in \mathbb{N}$

$$(m, m) \in R$$

$$|m - m| \leq 1$$

as  $m - m = 0$  &  $|0| = 0 \leq 1$  so  $R$  is reflexive on  $\mathbb{N}$ .  
(as  $m \neq 0$  because  $m \in \mathbb{N}$ )

(b)  $R$  is reflexive on  $\mathbb{Z} \times \mathbb{Z}$

$$\forall m \in \mathbb{Z} [(m, m) \in R]$$

supp  $m \in \mathbb{Z}$

$$(m, m) \in R$$

$$|m - m| \leq 1$$

No, because any -ve  $m$ , like  $-3$  or  $-1$ ,  $|m - m| = 0 \leq 1$  but  $-3$  is not in  $R$  as  $-3 \notin \mathbb{N}$ . Only  $m \in \mathbb{N}$  are allowed in  $R$  so reflexivity fails for domain  $\mathbb{Z}$ .

by definition  
of  $R$